

Project preview & user manual

Rea-Sixty

An open-source REAPER extension for the SSL UF8 and UC1.

Prepared for Solid State Logic as a courtesy preview ahead of any formal correspondence. Aim: transparency. What the project is, how it works, what ships today, what it asks of SSL.

Project	Rea-Sixty (repository: <i>reaper-uf8</i>)
Type	REAPER extension — C++, csurf_inst, libusb, vendored Dear ImGui (via RealmGui)
Hardware	SSL UF8 (VID 0x31E9 / PID 0x0021), SSL UC1 (PID 0x0023)
Platform	macOS (tested). Windows and Linux are on the roadmap, not yet built.
Source	github.com/acklin83/reaper-uf8
License	MIT — original code only; no SSL material is decompiled, reproduced, or redistributed.
Document date	2026-05-13

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1. What it is, in one paragraph

Rea-Sixty is an independent open-source REAPER extension that drives the SSL UF8 and UC1 directly. It claims the controllers' vendor-USB interface over libusb, replays the host-side initialisation sequence we observed SSL 360° emit, and runs the keepalive cycle that keeps the Plug-in Mixer rendering path alive. Everything the user sees on the hardware — track name, colour, fader, pan, sends, receives, automation state, focused-FX parameters, gain reduction — is read from REAPER's C API through a `csurf_inst`.

2. Why the project exists

The UF8's scribble strips can render DAW track colours, but only when the controller is in Plug-in Mixer Mode, which requires an SSL VST3 plug-in on every track that should appear. For sessions of 100+ tracks this is impractical. MCU and HUI carry no colour at all, so a "just talk MCU" workaround is impossible for the colour use case.

SSL 360° also holds the UF8's vendor-USB interface with an exclusive claim, so coexisting next to it is not possible. The only path was to take over 360°'s host-side responsibilities for REAPER specifically — which is what Rea-Sixty does.

3. How it works

On load, the extension enumerates SSL devices over libusb, claims their vendor interface, replays the captured init sequence, and starts a host thread that emits the Plug-in Mixer keepalive every 150 ms (both UF8 and UC1). Inbound vendor-USB frames are parsed into press / release / fader / V-Pot / knob events and dispatched through the REAPER C API. Outbound state is rendered to the controllers as scribble text, colour bars, LED frames, fader motor positions, gain-reduction LED ladders and VU meters.

Topology

```
REAPER <-> Rea-Sixty (csurf_inst) <-> UF8 / UC1 (vendor-USB, libusb)
```

Decoded protocol surface

- UF8 + UC1 frame format and checksum (FF <cmd> <len> <data> <chk>).
- UF8 init sequence and the 13 B / 64 B Plug-in-Mixer keepalive pair at 150 ms.
- UC1 init / keepalive (also 150 ms).
- Per-strip TFT colour command and the UF8's 16-entry palette. 11 indices have known RGB values; 5 are off / unmapped. Off-palette REAPER colours are matched by chromaticity to the closest known entry.
- All Plug-in Mixer scribble zones: track name, parameter label, value line, channel-strip type, O/PdB fader readout, channel number, colour bar.
- Per-strip SEL / MUTE / SOLO LED frames; transport LEDs; 12-segment metering frame.
- UF8 inbound button-ID map (per-strip + globals); fader 16-bit position and touch.
- UC1 inbound button + knob IDs (stable across plug-in contexts); GR meter (16-bit BE dB×10); VU bars; display zones.

4. What ships today

Status as of the document date. macOS only. Every entry below was verified directly in the source on origin/main — not inferred from plans or roadmap.

Capability	Notes
UF8 standalone replacement	<code>csurf_inst</code> with native init / keepalive / scribble / colour / LED / meter / fader motor / button routing. No SSL 360° required; no MCU bridge.
DAW-layer scribble-strip colours	Polled from <code>GetTrackColor()</code> and pushed on bank shift / theme change. Chromaticity-quantising palette match for off-palette colours.
16-bit fader resolution	Direct <code>CSurf_OnVolumeChange</code> ; no MCU 14-bit lossy stage.
UC1 parameter mirror	Physical knobs drive the focused track's SSL Channel Strip 2 / Bus Compressor 2 / 4K B / 4K E / 4K G / 360 Link in real time. Values mirrored back to the UC1 displays.
GR meter (UC1 + UF8)	Audio-driven, read via the PreSonus VST3 <code>GainReduction_dB</code> named-config-param. Same 30-sub-step index drives both the UC1 LED ladder and the UF8 GR byte, so the two hardware meters stay synchronised. Piecewise dB matches the SSL plug-in's 0–3–6–10–14–20 dB scale.
SSL Strip Mode	<code>ssl_strip_mode_toggle</code> (factory default: Shift+Plugin). Fader and V-Pot leave DAW track volume / pan and follow the focused SSL plug-in's output level / pan instead. With-GUI variant additionally opens / closes the plug-in's REAPER GUI.
UF8 Plug-in Deep Edit	<code>uf8_plugin_mode_toggle</code> — mutually exclusive with SSL Strip Mode; for deep editing inside a single plug-in instance.
Instance cycle	Channel encoder in Instance mode (or <code>instance_cycle</code> on any bound button) walks plug-in instances within the focused domain (Channel Strip or Bus Comp) on the selected track. Picks up UF8-only user-mapped plug-ins too.
Send / Receive layers	AllTracks-N (every strip shows send/receive index N for its track) and ThisTrack (8 strips show the focused track's first 8 sends/receives). Independent state for V-Pot vs Fader × Send vs Receive. LEDs, colours, names and mute follow the routed entity.
Folder Mode	Bindable surface filter: only top-level / depth-0 tracks fill the 8 strips. Long-press SEL (500 ms) on a folder-start track spills its children onto the strips to the right; long-press again to collapse.
Show Only Selected + Selection Sets	<code>show_only_selected</code> filters strips to live-selected tracks. 8 named slots (Selection Sets) recall via <code>selset_recall</code> param 1..8 — GUID-keyed so reorders / deletes don't corrupt them.
Bindings + Learn	Per-strip, transport, global and soft-keys. Modifier-aware dispatch (Shift / Fine, Norm, Rec, Auto). Multi-action chains with optional inter-step delays. Per-binding label, colour, brightness. Learn mode in the Settings screen.

Capability	Notes
12 user Soft-Key Banks + 6 SSL stock banks	Bindings.h kUserBankCount = 12, 8 slots per bank. The 6 SSL stock banks (V-POT, Bank 1..5) are addressable via softkey_bank_select with param 0..5.
FX Learn for any VST / JS / AU	GUID-keyed bindings survive FX-slot reorders. Param snapshot persisted, so a learned slot stays editable when no live FX instance is present. “Fill sequential (right)” on the right-click menu auto-fills consecutive parameters into the cells to the right of the current slot.
Virtual notch	Magnet behaviour at 0 dB / centre for EQ gains and pan: rotation snaps to neutral when crossing or entering the zone, then continues normally so 0.1 dB nudges still work.
Settings screen (in-REAPER window)	Dockable ImGui window with left-rail navigation. Sections: Device, Bindings, Soft-Key Banks, FX Learn, Modes, Selection Sets, About. CSI Import preview in the Bindings section for users migrating from CSI.
REAPER-theme follow	ThemeBridge polls REAPER's colour theme and re-tints the ImGui window live (light / dark switch follows immediately).

5. What is still under construction

Two areas are honest about not being done:

Plug-in Mixer pane

The in-REAPER window has a left-rail entry called “Mixer”. The rest of the rail (Device, Bindings, Soft-Key Banks, FX Learn, Modes, Selection Sets, About) is real and shipping. The Mixer pane itself is currently a scaffold: the window opens to it, but the contents are a placeholder reminding the developer what still needs to be drawn (channel-strip columns per track, Bus-Compressor rack, audio meters, fader / pan widgets). Source: `extension/src/MixerLayout.cpp`.

Windows and Linux ports

Capture and analysis tooling already runs on Windows; the Windows / Linux native build of the extension is on the roadmap and has not started.

6. Legal posture

Trademarks

“SSL”, “Solid State Logic”, “SSL 360°”, “UF8”, and “UC1” are trademarks of Solid State Logic Ltd. They appear in the project documentation solely to identify the hardware and software this extension interoperates with (nominative fair use). The project is not affiliated with, endorsed by, or sponsored by SSL.

Interoperability basis

Developed via independent passive observation of the USB wire protocol between legally purchased SSL UF8 / UC1 hardware and legally licensed SSL 360° software, for the sole purpose of interoperating with REAPER. No SSL code, firmware, binaries, or proprietary creative content is decompiled, reproduced, or redistributed. Cited authorities: EU Software Directive 2009/24/EC Art. 6 (interoperability exception); §69e UrhG (Germany); 17 USC §1201(f) (US interoperability exception).

No warranty

MIT licence, with an explicit notice in the README that running third-party firmware-level communication with SSL hardware may void the user's warranty.

7. Asks of SSL

Three asks, in plain language:

1. — Do you object to this being a public, open-source project? We'd rather hear it now than after the fact, and we're open to changing how the project is shaped.
2. — Would SSL be willing to share protocol documentation, under NDA if needed? Even partial reference would save us substantial capture-and-decode effort.
3. — If this could grow into something more formal (documented partner extension, listing alongside SSL's supported integrations), we'd be glad to talk.

Collaboration is the preferred outcome.

[USER MANUAL \(preview\)](#)

Rea-Sixty for REAPER

The user-facing manual that ships with the extension. Included so SSL can see what end-users are told to do.

8. Install

Prerequisites

- REAPER (any recent version).
- An SSL UF8, UC1, or both, connected by USB.
- macOS — Homebrew: `brew install libusb cmake pkg-config`.
- **SSL 360° must be quit before REAPER starts.** 360° holds the controller's vendor interface exclusively; Rea-Sixty cannot open it while 360° is running.

Build (macOS)

```
git clone https://github.com/acklin83/reaper-uf8.git
cd reaper-uf8/extension
cmake -B build
cmake --build build -j$(sysctl -n hw.ncpu)
```

Install into REAPER

Symlink (preferred during development — rebuilds are live):

```
ln -sf "$PWD/build/reaper_uf8.dylib" \
~/Library/Application\ Support/REAPER/UserPlugins/reaper_uf8.dylib
```

Or copy the dylib into the same folder. Restart REAPER.

Windows / Linux

Not yet. The capture and analysis tools run on Windows today; the extension itself is macOS-only at this time.

9. First start

- On load nothing visible appears inside REAPER itself; the extension is a control-surface plug-in with no menu entry.
- The UF8 leaves its “Awaiting Connection to SSL 360° Software” idle screen and enters Plug-in Mixer Mode.
- Scribble strips show the first eight tracks: track number, name, fader dB readout, V-Pot ring (Pan by default), and the DAW track-colour bar.
- The 12-segment meter on each strip follows REAPER's track meter.
- If a UC1 is connected, its knobs follow the focused track's SSL Channel Strip 2 / Bus Compressor 2 / 4K B / 4K E / 4K G. The GR LEDs follow the plug-in's gain reduction.
- Triggering the REAPER action *Rea-Sixty: Toggle Plugin Mixer Window* opens the on-screen window (left-rail navigation: Mixer / Device / Bindings / Soft-Key Banks / FX Learn / Modes / Selection Sets / About).

- If nothing happens, open REAPER's Console (View → Console). Rea-Sixty logs failures as `reaper_uf8: <reason>`. The most common reason is `SSL360Core` owns the device — quit `SSL 360°` and restart REAPER.

10. Buttons, faders, V-Pots

Factory defaults. Every entry below is a registered binding and can be re-bound in the Bindings section of the Settings screen.

Control	Default behaviour
Fader	REAPER track volume, 16-bit. Touch detected for automation latches.
V-Pot rotate	Pan by default. Flip swaps the V-Pot's target with the fader's. In SSL Strip Mode the V-Pot redirects to the focused SSL plug-in's pan.
V-Pot push	Reset target to default (pan = 0; FX param = plug-in default).
SOLO / CUT / SEL	Solo / Mute / Select on the strip's track. Single-press SEL is exclusive select.
BANK < / BANK >	Shift visible 8 by 8 (within whatever the current surface filter shows).
PAGE < / PAGE >	Soft-key bank page navigation (V-POT / Bank 1..5).
LAYER 1 / 2 / 3	Three bindable layers (Layer 2 and 3 start empty by factory default).
QUICK 1 / QUICK 2	Q1 focuses the Channel-Strip domain; Q2 focuses the Bus-Comp domain. Q3 unbound by default.
Transport	REAPER Play / Stop / Record / Rewind / Fast-forward.
360° key	Opens / closes Rea-Sixty's on-screen window (the same dockable ImGui window the toggle action targets).
UC1 knobs	Drive the focused track's Channel Strip / Bus Compressor in real time. The dedicated EQ / dynamics knobs are stably mapped; the top-centre V-Pot row is soft-mapped per focused plug-in variant.

11. Layers, modifiers, the Channel encoder

Layers

- Three layers (Layer 1, 2, 3). Each layer holds its own set of soft-key bindings; per-strip, transport and global bindings are layer-independent.
- Layer 1 ships with sensible defaults. Layers 2 and 3 start empty so users can build from scratch.
- `layer_select_1/2/3` jumps directly; `layer_select` with a param cycles.

Modifiers

- Shift / Fine and the SSL Norm / Rec / Auto row are dispatched as modifier flags. Each binding can match plain, +Shift, +Cmd, +Ctrl etc.; the matcher prefers the most specific binding present.
- On a button release the binding fires under whatever modifier was held at press time, so a modifier released early still produces the “intended” release event.

- LED preview: dim when the modifier-specific binding is inactive, bright when held and active. Per-binding colour overrides take precedence over the stock SSL colour.

Channel encoder modes

Mode	Behaviour
Nav (default)	Track select — rotate to move the selection.
Nudge	Nudges the playhead by the REAPER nudge amount.
Focus	Mouse-scroll-wheel emulation at the cursor.
Instance	Cycles plug-in instances within the focused domain on the selected track.

The encoder push (channel-encoder press) resets the mode to Nav. Bindings `encoder_nav`, `encoder_nudge`, `encoder_focus`, `encoder_instance` set the mode directly from any button.

12. Surface filters: Folder Mode, Show Only Selected, Selection Sets

These three are surface filters — they change which tracks fill the 8 strips without changing the project. They stack with the bank / channel buttons.

Folder Mode

- Bind any button to `folder_mode`. Toggle on: only folder-start tracks (`I_FOLDERDEPTH >= 1`) and depth-0 children fill the strips.
- Long-press SEL on a folder-start track (500 ms) spills that folder's children onto the strips immediately to its right. Long-press again to collapse.
- REAPER's MCP folder-collapse state is left alone — Folder Mode is a Rea-Sixty-only surface filter, the TCP view does not change.
- Persists across REAPER restarts via the `ReaSIXTY/folderMode ExtState`.

Show Only Selected

Bindable filter: only currently-selected tracks fill the strips. Useful when you want to focus a few specific tracks without committing to a saved set.

Selection Sets

Eight saved slots, recalled via `selset_recall` with param 1..8. Slots are GUID-keyed (track GUIDs, not indices) so reorders and deletes don't corrupt them. The Selection Sets tab of the Settings screen manages save / recall / clear.

13. Send / Receive routing

Rea-Sixty has four independent routing states — V-Pot vs Fader, each × Send vs Receive. They behave consistently:

- **AllTracks-N** (param 1..8) — every visible strip shows send (or receive) index *N* for its own track. Default factory binding: the Send/Plugin button row above the V-Pots.
- **ThisTrack** — the 8 strips show the focused track's first 8 sends (or receives). Bound to `send_this` / `recv_this`.
- Within a domain (e.g. V-Pots showing sends) AllTracks-N and ThisTrack are mutually exclusive.

- Send and Receive on the same physical output (V-Pot vs Fader) are mutually exclusive — AllTracksReceive3 on the V-Pots cancels any active Send mode on the V-Pots.
- V-Pot and Fader stay independent: V-Pots can show Sends while Faders still drive track volume.
- LEDs, colours, names, and CUT/Mute all follow the routed entity rather than the host track while a routing mode is active.

14. SSL Strip Mode and Instance Cycle

SSL Strip Mode

Two builtins flip `g_pluginFaderMode`:

- `ssl_strip_mode_toggle` — UF8 fader and V-Pot leave DAW track volume / pan and follow the focused SSL plug-in's output level / pan instead.
- `ssl_strip_mode_toggle_with_gui` — same, plus opens / closes the focused plug-in's REAPER GUI on the main thread. Factory-bound to Shift+Plugin.
- Mutually exclusive with `uf8_plugin_mode_toggle` (UF8 Plug-in Deep Edit). Turning one on forces the other off.
- Persists across restarts via the `ReaSixty/pluginFaderMode ExtState`.

Instance Cycle

- Channel encoder set to Instance mode (or any button bound to `instance_cycle`) walks plug-in instances on the focused track within the active domain.
- Channel-Strip domain: cycles through CS2 / 4K B / 4K E / 4K G / 360 Link variants present on the track.
- Bus-Comp domain: cycles through every Bus Comp instance on the track.
- User-mapped UF8-only plug-ins (entered via FX Learn) participate in the cycle alongside the stock SSL ones.

15. FX Learn

FX Learn lets any VST / JS / AU plug-in be controlled from the UF8 (and selectively from the UC1's soft-mapped knob row). Open **Settings** → **FX Learn**:

- Add an FX from the catalogue (or pick one from the focused track). A schematic UF8 mockup appears with bindable cells.
- Arm a cell, then touch the parameter inside the plug-in's REAPER window. The binding records the plug-in's GUID and the parameter index; FX-slot reorders don't break it.
- **Fill sequential (right)**: right-click a learned cell whose parameter name contains a digit (e.g. "Band 1 Freq"). The remaining cells to the right auto-fill with the next consecutive parameters of the same plug-in.
- Each entry carries a label, colour, brightness, and a UF8-only / UC1-shared mode flag.
- Param values are formatted via `TrackFX_GetFormattedParamValue` — the plug-in's own formatter is used ("−3.5 dB", "42 %"); no per-plug-in formatter code in Rea-Sixty.
- Param snapshots are persisted so a learned slot stays editable when the live FX instance isn't present.

16. Bindings file & CSI import

Bindings are stored in a JSON file under REAPER's resource directory. The Settings screen edits this file; Learn-Mode edits it; the extension watches the file's mtime and reloads on change with no REAPER restart needed.

Location

OS	Path
macOS	~/Library/Application Support/REAPER/rea_sixty/bindings.json
Windows (future)	%APPDATA%\REAPER\rea_sixty\bindings.json
Linux (future)	~/.config/REAPER/rea_sixty/bindings.json

Binding types

- **reaper_action** — dispatches a REAPER action (numeric ID or named SWS / ReaPack / user-macro).
- **builtin** — surface behaviour: bank nav, layer / V-Pot mode, flip, folder/show-only-selected, send/receive routing, instance cycle, SSL Strip Mode toggle, modifier flags, encoder modes, brightness, etc. (65 builtins total).
- **track_target** — per-strip standard target (select, mute, solo, rec_arm, ...).
- **fx_param** — bind a V-Pot or soft-key to an FX parameter; GUID-keyed.
- Multi-action chains: any slot can carry up to N steps with optional inter-step delays; single-step chains run synchronously, multi-step ones are queued and drained from the main-thread timer.

CSI import

Users migrating from a CSI surface configuration can point Rea-Sixty at the CSI Surface directory (defaults to ~/Library/Application Support/REAPER/CSI/Surfaces/SSLUF8). The Bindings section reads CSI's Home.zon and previews the resulting bindings before they're committed.

17. Troubleshooting

Symptom	Likely cause / fix
UF8 stays on “Awaiting Connection to SSL 360° Software”.	SSL 360° is still running. Quit it, restart REAPER.
Console: SSL360Core owns the device.	Same cause — the vendor interface is exclusively claimed.
Scribble strips light up but stay blank.	Init replayed but the 150 ms Plug-in-Mixer keepalive isn't reaching the device. Check the Console for libusb errors, reseal the USB cable, restart REAPER.
Track colour on the strip is “close but not exact”.	REAPER's colour is being matched by chromaticity to the closest of the UF8's 11 known palette entries (16 total, 5 unmapped). The match is deliberately conservative.

Symptom	Likely cause / fix
UC1 GR LEDs sit at zero with audio playing.	The focused track has no SSL Channel Strip / Bus Comp on it, or its plug-in does not expose the PreSonus <code>GainReduction_dB</code> named-config-param. Re-focus the track in question.
The Mixer pane in the on-screen window shows a “scaffold” message.	It is a scaffold today — the channel-strip + Bus-Comp rendering is on the roadmap. Settings (Device, Bindings, Soft-Key Banks, FX Learn, Modes, Selection Sets, About) all work in the same window.

Issues that aren't on the list:

github.com/acklin83/reaper-uf8/issues

18. Uninstall

Delete the dylib (or its symlink) and restart REAPER:

```
rm ~/Library/Application\ Support/REAPER/UserPlugins/reaper_uf8.dylib
```

User data — bindings, soft-key banks, FX-Learn catalogue — lives in `~/Library/Application Support/REAPER/rea_sixty/`. Removing the dylib leaves that folder intact. Delete the folder by hand to wipe every Rea-Sixty trace.

End of preview

Source of truth: github.com/acklin83/reaper-uf8. Questions, objections, or an opening to collaborate are welcome — see section 7.